

Name of Student: Shreyas Kulkarni	PRN:25070122225
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 1	
TITLE : Java Environment Setup and Basic Program Execution	

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

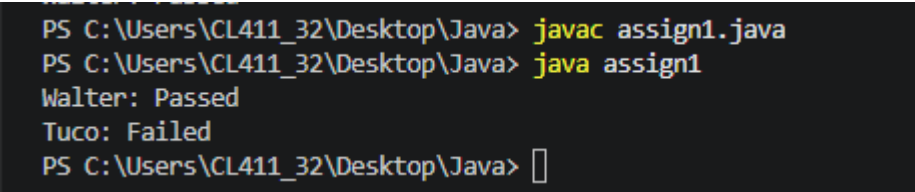
Programme:

```
class Student{  
    // variables  
  
    String name;  
  
    int score;  
  
  
    // constructor  
    public Student(String name, int score){  
        this.name=name;  
        this.score=score;  
    }  
  
  
    // method
```

```
public void printResult(){  
    if(score>=40) System.out.println(name + ": Passed");  
    else System.out.println(name + ": Failed");  
}  
}
```

```
public class assign1{  
    public static void main(String args[])  
    {  
        Student s1 = new Student("Walter", 99);  
        Student s2 = new Student("Tuco", 33);  
        s1.printResult();  
        s2.printResult();  
    }  
}
```

Output



```
PS C:\Users\CL411_32\Desktop\Java> javac assign1.java  
PS C:\Users\CL411_32\Desktop\Java> java assign1  
Walter: Passed  
Tuco: Failed  
PS C:\Users\CL411_32\Desktop\Java> █
```

Conclusion

This application demonstrates the core fundamentals of Java software development - setting up the runtime environment (JDK, JRE, JVM). By combining variables to store state, constructors to initialize objects safely, and methods to drive application behavior, we created a modular and structured program.

Questions

1. Explain the main concept used in Java environment setup, variables and methods.

Ans: Java environment is used to run, variables to store and methods to perform actions.

Java Environment Setup:

- **JDK (Java Development Kit):** The full software development kit containing tools needed to write, compile, and debug Java programs (includes `javac`, `java`, libraries, and JRE).
- **JRE (Java Runtime Environment):** The runtime package providing the class libraries and resources necessary to execute Java compiled code (includes JVM).
- **JVM (Java Virtual Machine):** The abstract engine that executes Java bytecode (`.class` files) line-by-line, translating it into machine-native instructions.

Variables: These are the containers used to store data values in memory.

Methods: A block of code used to perform certain operations. This is very helpful in terms of code reusability as we don't have to write the same code each and every time.

2. What are the advantages of using Java variables and methods?

Ans: **Variables** store data and give it meaning.

Methods perform actions and reuse logic.

- **Variables (Data):** Catch errors early with strict types, make code readable, and let you update values in one place.
- **Methods (Actions):** Prevent repeating code, break big problems into small tasks, and hide complex logic behind a simple name.

Together, variables hold your program's state and methods drive its behavior.

3. How can this implementation be improved?

Ans:

1. Make variables **private** (Encapsulation)

Currently, anyone can directly change `score` from outside the class (e.g., `s1.score = -50`). Making variables **private** prevents unauthorized or accidental changes.

2. Validate input in the Constructor

Add a simple check so bad data (like negative scores or scores over 100) gets blocked right away.

3. Return values instead of printing directly

Having a method return a boolean or string makes your code much more flexible—you can show the result in a console, a mobile app, or a web page.

4. Mark unchangeable fields as **final**

Since a student's name usually doesn't change after creation, making the name final prevents it from being reassigned later.

4. What are the real-world applications?

Ans:

Real-World Applications

1. Banking and Financial Systems (Core Banking / Fintech)

- Real Application: Processing savings, checking, and loan accounts in production platforms.

2. E-Commerce Platforms (Shopping Carts & Orders)

- Real Application: Platforms like Amazon or Shopify managing customer order processing.

3. Enterprise Resource Planning (ERP) & Inventory Systems

- Real Application: Warehouse tracking systems maintaining stock balances.

4. RESTful Web Services & Spring Boot Applications

- Real Application: Microservices serving JSON responses to web and mobile apps.